

Two legs of degree two: the extension-classifying cocycle H^2 and the semi-abelian homology H_2 of skew braces, and the open bridge between them

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Abstract

Two “second (co)homologies” now circulate in skew-brace theory, and it is tempting to read one as a duplicate — or a scoop — of the other. They are not the same invariant seen twice. The first is a *cohomology* H^2 that classifies extensions of a skew brace by an ideal: its coefficient is a *module*, and its data include a *residue-dependent* twist coming from the multiplicative action. The second is a *homology* H_2 of Barr–Beck/Hopf-formula type in the semi-abelian category of skew braces: its coefficient is a choice of *reflector*, its action is *fixed*, and it is a Baer invariant of a single object rather than a classifier of extensions. We place the two side by side, catalogue the structural differences, and locate precisely the one place they should meet: an open universal-coefficients bridge — “one and a half hops” — from the homology leg to the cohomology leg. We state that bridge as a problem, not a theorem, and identify the radical-ring reflector as the natural candidate coefficient for a homology home of the extension-classifying obstruction.

1 Introduction

The problem: “degree two” is ambiguous. Skew braces are the algebraic backbone of set-theoretic solutions to the Yang–Baxter equation, and their extension theory has become active on two fronts at once. Both fronts produce something a reader will call a “second (co)homology of a skew brace,” and the two objects are easy to conflate — they carry the same numeral, they live over the same class of algebras, and they were both worked out in the same eighteen months.

On the first front, Rathee and Yadav [1] classify skew-brace extensions $0 \rightarrow D \rightarrow G \rightarrow H \rightarrow 0$ by a second cohomology $H_{\text{sb}}^2(H, D)$ whose cocycles are *pairs* (β, τ) — an additive 2-cocycle and a multiplicative 2-cocycle coupled by a compatibility identity. On the second front, Gran, Letourmy and Vendramin [2] develop a *homology* of skew braces of Barr–Beck type, presented by relative Hopf formulae in the semi-abelian category SKB and yielding a Stallings–Stammbach five-term exact sequence.

The gap. No account has yet said plainly how these two degree-two invariants relate. The silence invites two opposite errors. One is to assume they coincide — that the homology H_2 *is* the cohomology H^2 dualised, so that either program subsumes the other. The other is to assume they are simply unrelated. Both errors matter here concretely: the present author’s obstruction-theoretic results on skew-brace decomposition — the bilinear class $[\Omega]$ and the residue-dependent solvability obstruction o_ν — live squarely on the cohomology leg, and it is a genuine question whether the homology leg scoops them, duplicates them, or neighbours them.

The result of this note. It neighbours them. We give a side-by-side dictionary (Table 1) of the two legs and argue that they differ in three structural invariants that no change of bookkeeping can reconcile:

- (i) **Coefficients.** The cohomology leg takes a *module* D ; the homology leg takes a *reflector* $X \in \{\text{Grp}, \text{RadRng}, \text{BR}, \text{Ab}\}$ into a subvariety.
- (ii) **Residue.** The cohomology leg is *residue-dependent*: its data include the multiplicative action ν and the $\circ$ -conjugation twist. The homology leg is *fixed-action*: the actions factor through the abelianisation and there is no residue datum at all.
- (iii) **What is classified.** The cohomology leg *classifies extensions*; the homology leg is a *Baer invariant of one object*, and it treats non-abelian kernels only for *central* extensions.

Having separated them, we then locate the single place they *should* meet: a universal-coefficients passage from H_2 to H^2 , scaffolded by the five-term sequence of [2] but left open there. We state it as Problem 3 and no more — this is a positioning note, not a claim.

Why it matters. For the compositional-correctness programme the payoff is a clean map of the territory: the $[\Omega]/o_\nu$ package is not scooped by the semi-abelian homology, because that homology computes a different invariant of a different flavour; and the two nonetheless share a precise, buildable target. Naming that target is the point.

Outline. Section 2 fixes the minimal vocabulary. Section 3 is the cohomology leg (Rathee–Yadav, and the author’s $[\Omega]$ and o_ν). Section 4 is the homology leg (Gran–Letourmy–Vendramin). Section 5 is the dictionary. Section 6 states the open bridge. Section 7 draws the consequence for the programme.

A typographic warning. Throughout, H^2 (superscript) is always the *cohomology* of Section 3 and H_2 (subscript) always the *homology* of Section 4. The distinction between the two legs is exactly the distinction between the two positions of the “2”.

2 Preliminaries

A *skew brace* is a set A with two group structures $(A, +)$ and $(A, \circ)$, sharing a unit 0, related by the compatibility

$$a \circ (b + c) = (a \circ b) - a + (a \circ c) \quad (a, b, c \in A).$$

Writing $\lambda_a(b) := -a + (a \circ b)$ gives the λ -action $\lambda: (A, \circ) \rightarrow \text{Aut}(A, +)$. An *ideal* $D \trianglelefteq A$ is a normal subgroup of both $(A, +)$ and $(A, \circ)$ that is stable under λ ; it is exactly the kernel of a skew-brace morphism, so ideals index quotients. A skew brace is a *trivial brace* when $\circ = +$ (equivalently $\lambda \equiv \text{id}$). The *socle* is $\text{Soc}(A) = \ker \lambda \cap Z(A, +)$. Skew braces and their morphisms form a variety SKB, which is *semi-abelian* [2].

Fix a skew-brace *extension* of a base H by an ideal D ,

$$0 \longrightarrow D \longrightarrow G \longrightarrow H \longrightarrow 0, \tag{1}$$

and a normalised set-theoretic section $s: H \rightarrow G$ with $s(0) = 0$. Following [1] one reads off the *good triplet*

$$\nu: (H, \circ) \rightarrow \text{Aut}(D, +), \quad \mu: (H, +) \rightarrow \text{Aut}(D, +), \quad \sigma: (H, \circ) \rightarrow \text{Aut}(D, +),$$

by $\nu_h = \lambda_{s(h)}|_D$, $\mu_h(x) = -s(h) + x + s(h)$, and $\sigma_h(x) = s(h)^{-1} \circ x \circ s(h)$ (the last is the $\circ$ -conjugation, which we also write $\widehat{\sigma}_h$). The triplet is independent of the choice of section. We call ν the *residue* of the extension; it is the datum by which the multiplicative structure of G acts on the additive structure of D , and it is precisely what the homology leg will not see.

3 The cohomology leg: extensions and their obstructions

3.1 The Rathee–Yadav classification

The extensions (1) realising a fixed good triplet are classified by a second cohomology $H_{\text{Sb}}^2(H, D)$ [1]. A cocycle is a *pair*

$$(\beta, \tau), \quad \beta: H \times H \rightarrow D \text{ (additive 2-cocycle),} \quad \tau: H \times H \rightarrow D \text{ (multiplicative 2-cocycle),}$$

subject to a compatibility identity coupling β and τ through the actions λ, ν, μ (Rathee–Yadav’s equation (3.10)). The class $[(\beta, \tau)]$ vanishes iff the extension splits, i.e. iff G decomposes as an internal Zappa–Szép-type product of H and D . The coefficient is the *module* D together with its actions; the theory in [1] is developed for D a *trivial brace* kernel.

The point to hold onto is that a single degree-two class already *carries two group Schreier classes inside it*: forgetting τ leaves an additive class, forgetting β leaves a multiplicative one, and the compatibility identity binds them.

3.2 The bilinear obstruction $[\Omega]$

For the decomposition problem “does G split as $(G/D) \bowtie D$ along an ideal D ?” the obstruction is a class $[\Omega]$, and it *is* the Rathee–Yadav class $[(\beta, \tau)]$ in the trivial-kernel case [1]; this identification has been peer-reviewed within the group. Its structure is genuinely bilinear, and the following facts pin down how:

- The *additive forgetful map* $\varphi_+ : [(\beta, \tau)] \mapsto [\beta]$ into the group cohomology $H_{\text{Grp}}^2((H, +), D; \mu)$ is well defined, with no extra hypothesis: a change of section shifts β by the standard additive 2-coboundary, independently of τ . (The *multiplicative forgetful map* $\varphi_\circ : [(\beta, \tau)] \mapsto [\tau]$ is Rathee–Yadav’s own.)
- The *product map* $\Phi = (\varphi_\circ, \varphi_+)$ into $H_{\text{Grp}}^2((H, \circ), D) \times H_{\text{Grp}}^2((H, +), D; \mu)$ *fails to be injective*: on the 4×4 witness $H = \mathbb{Z}/2 \times \mathbb{Z}/4$ there is a class with $[\Omega] \neq 0$ but $\Phi([\Omega]) = (0, 0)$. The obstruction is therefore irreducibly bilinear — it lives in the coupling and is strictly stronger than either group Schreier class.

Both the descent of φ_+ and the Φ -non-injectivity are proved, and the crux lemma (the section-change computation) is machine-checked in Lean 4.

3.3 The residue-dependent solvability obstruction o_ν

The forgetful maps raise the reverse question: which additive classes $[\beta] \in H_{\text{Grp}}^2((H, +), D; \mu)$ actually *lift* to a full skew-brace cocycle inducing the fixed triplet — that is, what is $\text{im } \varphi_+$? The answer is an obstruction o_ν , and its defining feature is that it is *residue-dependent*: the additive class $[\beta]$, the residue ν , and the multiplicative twist are *coupled*, not independent.

Theorem 1 (trivial-brace-kernel regime; proved). *In the Rathee–Yadav (trivial-brace-kernel) setting, o_ν is a well-defined group homomorphism*

$$o_\nu: H_{\text{Grp}}^2((H, +), D; \mu) \longrightarrow C^3/A(Z_\circ^2), \quad [\beta] \mapsto [Q\beta],$$

killing the additive coboundaries, and

$$\text{im } \varphi_+ = \ker o_\nu.$$

Equivalently, an additive class lifts to a skew-brace cocycle with the given triplet iff $o_\nu([\beta]) = 0$.

The proof is structural: the coupling operators are $\mathbb{Z}$ -linear, a “linchpin” coboundary identity shows o_ν kills B_+^2 , and Rathee–Yadav’s cocycle characterisation supplies the surjection onto genuine cocycles as a cited black box. The $|G| = 8$ collapsed slice ($H = \mathbb{Z}/2$, $D = \mathbb{Z}/4$) is formalised sorry-free in Lean 4. Two honesty flags travel with the theorem: (a) it is proved *in the trivial-brace-kernel regime*, resting on [1]’s characterisation as a premise; and (b) beyond that regime the picture is sharper but not yet fully proved, as follows.

Remark 2 (non-trivial kernel; proved at $H = \mathbb{Z}/2$, computed in general). When D is a *non-trivial* brace, the correct home of the multiplicative datum is the group $(D, \circ)$ twisted by the $\circ$ -conjugation $\widehat{\sigma}$ — not the additive $(D, +)$ — and there is an intrinsic *defect* $\delta_h = \nu_h \widehat{\sigma}_h - \text{id}$ measuring the failure of the trivial-kernel collapse. For $H = \mathbb{Z}/2$ this yields a proved “three-way lock” $(\lambda_x^D - 1)b = 2(\nu_1 \widehat{\sigma}_1(x) - x)$ (verified on all 112 genuine kernels), and $\text{im } \varphi_+ = o_\nu^{-1}(0)$ becomes an *affine* coset: it is a torsor with no basepoint exactly when $L := \text{im } \lambda^D \subsetneq \text{Aut}(D)$ and $\nu \widehat{\sigma}$ is non-sole. The general- H form of the obstruction is validated at operator level (16/16 triplets for $H = \mathbb{Z}/2$, exact assembly 33/33) but its closed-form offset for general H is *computed*, not yet proved — one symbolic step remains open.

3.4 The character of the leg

Three features recur. The coefficient is a *module* D . The theory is *residue-dependent*: ν and the $\circ$ -twist are part of the data, and o_ν exists precisely to measure their coupling to $[\beta]$. And the invariant *classifies extensions*: a class is an extension up to equivalence, and it vanishes iff the extension splits. Every one of these three is reversed on the other leg.

4 The homology leg: Baer invariants of one object

4.1 Barr–Beck homology via Hopf formulae

Gran, Letourmy and Vendramin [2] compute a *homology* of skew braces in the semi-abelian category SKB. It is comonadic (Barr–Beck) homology, made explicit by a *relative Hopf formula*: for a free presentation $0 \rightarrow K \rightarrow F \rightarrow B \rightarrow 0$ and a reflector $F_X: \text{SKB} \rightarrow X$ onto a subvariety X ,

$$H_2(B, F_X) \cong \frac{K \cap [F, F]_X}{[K, F]_X}, \quad H_1(B, F_X) \cong \frac{B}{[B, B]_X}, \quad (2)$$

where $[\cdot, \cdot]_X$ is the X -relative commutator. The genuinely new algebraic object is the *radicalator* $[I, A]_{\text{RadRng}}$, the relative commutator for the reflector onto radical rings; the Huq and Smith commutators coincide with the abelian commutator $[\cdot, \cdot]_{\text{Ab}}$. The main extension-level output is a Stallings–Stammbach *five-term exact sequence* for a short exact sequence of skew braces.

4.2 The character of the leg

The coefficient here is not a module: it is a *choice of reflector* $X \in \{\text{Grp}, \text{RadRng}, \text{BR}, \text{Ab}\}$, i.e. a choice of “abelianisation direction” into a subvariety. The theory is *fixed-action*: the λ - and ρ -actions factor through the additive abelianisation A_+/J_+ , so there is *no residue datum* ν — the very thing the cohomology leg is built around is quotiented away here by construction. And the invariant is a *Baer invariant of one object* B : $H_2(B, F_X)$ is an invariant of B (and X), computed from any free presentation, not a set of extensions of B by a prescribed module. Non-abelian kernels are treated only for *central* extensions, and the paper carries no numerical examples — a size-24 database counterexample is invoked only to *separate* commutators, not to compute a class.

5 The dictionary

Table 1 is the centre of this note. Read down each column and the two legs are seen to disagree on every structural axis; read across each row and one sees why neither can be a relabelling of the other.

	Cohomology leg (§3)	Homology leg (§4)
Invariant	$H_{\text{Sb}}^2(H, D)$ (superscript)	$H_2(B, F_X)$ (subscript)
Type	extension- <i>classifying</i>	<i>Baer invariant</i> of one object
Coefficient	a <i>module</i> D (with actions)	a <i>reflector</i> $X \in \{\text{Grp}, \text{RadRng}, \text{BR}, \text{Ab}\}$
Cocycle / formula	pair (β, τ) + compatibility (3.10)	Hopf formula $(K \cap [F, F]_X)/[K, F]_X$
Action	<i>residue-dependent</i> : $\nu, \hat{\sigma}$ are data	<i>fixed</i> : actions factor through A_+/J_+
Residue ν ?	yes — o_ν measures its coupling	no — quotiented away
Non-abelian kernel	general ideal D	<i>central</i> extensions only
Vanishing means	extension splits	(leg-specific; not a splitting criterion)
Extension output	the classification itself	Stallings–Stammbach 5-term sequence
Examples	explicit $(\mathbb{Z}/2 \times \mathbb{Z}/4, \mathbb{Z}/8, \dots)$	none (separating counterexample only)
Author’s stake	$[\Omega]$, $\varphi_\pm$, o_ν live here	(neighbour; no overlap)
Sources	Rathee–Yadav [1]	Gran–Letourmy–Vendramin [2]

Table 1: The two legs of degree two. The single class H_{Sb}^2 classifies extensions with a residue-dependent module coefficient; the homology H_2 is a fixed-action Baer invariant with a reflector coefficient. No column agrees.

The decisive rows are *coefficient*, *residue*, and *type*. A reflector is not a module and a module is not a reflector; a residue-dependent theory cannot be a fixed-action one; and a classifier of

extensions is not a Baer invariant of a single object. These are not bookkeeping differences that a dualisation would erase.

6 The open bridge

And yet the two legs are not strangers. Degree-two homology and degree-two cohomology are related, in ordinary algebra, by universal coefficients: $H^2(-; M)$ is built from $H_2(-)$ by dualising into M , modulo an Ext term. The question is whether the semi-abelian H_2 of [2] feeds the extension-classifying H^2 of [1] the same way. The honest description of the distance is *one and a half hops*:

Hop 1. From the homology $H_2(B, F_X)$ to a derived functor of abelianisation. This is exactly what the Hopf formula (2) provides, and [2]’s five-term sequence is the scaffold linking H_2 , H_1 , and the abelianised kernel across an extension.

Half a hop. From that derived functor, *dualise / apply universal coefficients* into a module coefficient D to reach a cocycle H^2 with module coefficients. This is the step that is *not* carried out in [2] — it is left open there — and it is only “half” a hop because the target is not the naïve $H^2(-; D)$ but the residue-carrying $H_{\text{sb}}^2(H, D)$, and the residue ν must be *reintroduced* on the cohomology side after the homology side has quotiented it away.

Problem 3 (the bridge). Construct a universal-coefficients passage from the semi-abelian homology $H_2(B, F_X)$ of [2] to the extension-classifying cohomology $H_{\text{sb}}^2(H, D)$ of [1], with the five-term sequence of [2] as scaffold. In particular, determine which reflector X and which coefficient recover the residue-dependent data $(\nu, \hat{\sigma})$ of the cohomology leg, so that the obstructions $[\Omega]$ and o_ν acquire a homological description.

We do not claim this bridge; we state it. Two remarks constrain the search. First, because the homology leg is fixed-action, any bridge must *reintroduce* the residue — so a plain dualisation cannot suffice, and the interesting content is exactly the recovery of ν . Second, the radical-ring reflector is the natural candidate coefficient: the radicalator $[I, A]_{\text{RadRng}}$ and RadRng -centrality are the part of [2] that sees multiplicative (as opposed to purely additive) structure, which is where a homological home for the bilinear, $\circ$ -twisted obstruction $[\Omega]$ would have to live. We flag $X = \text{RadRng}$ as the candidate $[\Omega]$ -analogue home and leave the construction open.

7 Consequence for the programme

The extension-classifying obstructions $[\Omega]$ and o_ν have a published homology *neighbour* — the semi-abelian H_2 of Gran–Letourmy–Vendramin — which they neither scoop nor duplicate. The two compute different invariants of different type over the same algebras: a residue-dependent, module-coefficient classifier of extensions on one side, a fixed-action, reflector-coefficient Baer invariant on the other. For the compositional-correctness programme this is the clean statement we wanted: the cohomological work is not pre-empted by the homological work, and the honest joint target is the bridge of Problem 3, whose natural coefficient we conjecture to be the radical-ring reflector. Building that bridge — reintroducing the residue on the homology side — is the next question, and it is a question about both legs at once.

Provenance and scope. The Rathee–Yadav classification and their multiplicative forgetful map are cited, not reproved. The identification $[\Omega] = [(\beta, \tau)]$ is peer-reviewed within the group; the descent of φ_+ and the non-injectivity of Φ are proved and Lean-checked. Theorem 1 is proved in the

trivial-brace-kernel regime, resting on [1]’s cocycle characterisation as a premise; the non-trivial-kernel sharpening of Remark 2 is proved at $H = \mathbb{Z}/2$ and computed in general, with one symbolic step open. The facts attributed to [2] are from a full-text reading of that paper. The bridge of Section 6 is open and is not claimed. Gran–Letourmy–Vendramin themselves cite Rathee–Yadav’s earlier 2024 cohomology paper (a distinct paper from the extension paper [1] used here), a modest pre-existing bibliographic contact point between the semi-abelian and skew-brace traditions — but not the bridge, which remains to be built.

References

- [1] M. Rathee and M. K. Yadav, *Skew brace extensions, second cohomology and complements*, arXiv:2601.12371.
- [2] M. Gran, T. Letourmy and L. Vendramin, *Hopf formulae for the homology of skew braces*, J. Pure Appl. Algebra **229** (2025), no. 11, 108085; arXiv:2409.18056.